Machine Learning for Molecular Solubility Prediction

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Objective

The goal of this independent project is to learn the complete machine-learning workflow through the prediction of aqueous molecular solubility using the ESOL dataset [1]. The target is the experimentally measured logarithmic solubility,

$$y = \log_{10}(S),$$

where S is measured in mol/L. ESOL contains 1,128 molecules and is small enough for rapid experimentation while still representing a nontrivial molecular property-prediction problem.

The project has progressively examined three questions:

1. How much can standard regression and tree-based models learn from simple molecular descriptors?
2. How strongly does the choice of molecular representation affect prediction?
3. How does model performance change when evaluation is made more chemically realistic?

1. Initial Modeling

The first models used six numerical descriptors supplied with the dataset: Molecular Weight, Polar Surface Area, Number of Rings, Number of Rotatable Bonds, Number of H-Bond Donors, and Minimum Degree.

The workflow consisted of exploratory data analysis, an 80/20 train–test split, 5-fold cross-validation, hyperparameter optimization, and final evaluation using RMSE, MAE, and R^2.

Model	**Test RMSE**	**Test R^2**
Linear Regression	1.199	0.696
Polynomial Ridge	1.057	0.764
Random Forest	0.875	0.838
Gradient Boosting	0.876	0.838
XGBoost	0.874	0.838

The similar performance of the three tree ensembles suggested that the main limitation was no longer the optimization algorithm, but the information content of the six features. Tree-based methods clearly improved over the linear models, indicating important nonlinear relationships and feature interactions.

2. Chemistry-Informed Feature Engineering

The next experiment introduced the RDKit Crippen logP descriptor, which measures hydrophobicity. Keeping the train–test split and model fixed allowed the effect of this single new feature to be isolated.

For XGBoost,

$$\text{RMSE}: \quad 0.874 \to 0.741.$$

Thus, adding one physically motivated descriptor reduced random-split test RMSE by approximately 15%. SHAP analysis subsequently identified logP as the dominant predictive feature, followed by Polar Surface Area, Molecular Weight, and Number of Rings.

This result demonstrated an important lesson of the project:

$$\boxed{\text{better molecular representation can matter more than additional hyperparameter tuning.}}$$

3. Chemistry-Aware Evaluation

Random train–test splitting can place structurally similar molecules in both sets. To evaluate generalization to unfamiliar molecular families, molecules were therefore grouped using *Bemis–Murcko* scaffolds [2].

With the 7-feature XGBoost model,

$$\mathrm{RMSE}_{\mathrm{random}} = 0.741, \qquad \mathrm{RMSE}_{\mathrm{scaffold}} = 0.940.$$

The higher scaffold-split error shows that predicting molecules with unfamiliar structural frameworks is substantially harder than interpolating among chemically similar examples.

The feature representation was then expanded to approximately 199 RDKit physicochemical descriptors. For this representation:

Evaluation	**CV RMSE**	**Test RMSE**
Random split + random KFold	0.588	0.611
Scaffold split + random KFold	0.548	0.889
Scaffold split + GroupKFold	0.812	0.886

The large mismatch between random-CV RMSE and scaffold-test RMSE disappeared when the cross-validation procedure was also made scaffold-aware using `GroupKFold`. This showed that the validation strategy must reflect the same generalization problem as the final test set.

The current most defensible performance estimate is therefore approximately

$$\boxed{\mathrm{RMSE} \approx 0.89}$$

for XGBoost using the full RDKit descriptor representation under scaffold-aware training and evaluation.

4. Graph Neural Network Baseline

A baseline two-layer Graph Convolutional Network (GCN) was also trained directly from molecular graphs using PyTorch Geometric.

Under the same scaffold split,

$$\mathrm{RMSE}_{\mathrm{GCN}} = 1.220, \qquad \mathrm{RMSE}_{\mathrm{XGBoost}} = 0.940$$

for the earlier 7-feature comparison.

The GCN therefore did not outperform the engineered-feature model. Possible reasons include the small training set, limited use of bond information in the vanilla GCN architecture, and the use of simple mean pooling. This demonstrated that learned graph representations are not automatically superior in a small-data regime.

5. Current Experiment: Morgan Fingerprints

The current experiment changes the molecular representation again while keeping the downstream XGBoost/scaffold-aware evaluation pipeline fixed.

Three representations are being compared:

1. 199 RDKit descriptors,
2. 2048-bit Morgan fingerprint ($r = 2$),
3. RDKit descriptors + Morgan fingerprint ($\sim$2247 features).

Morgan fingerprints encode local molecular environments as binary features. Unlike RDKit descriptors, which describe global physicochemical properties, Morgan fingerprints provide explicit information about molecular substructures.

The key question is therefore:

$$\boxed{\text{Do local structural motifs provide complementary information beyond global physicochemical descriptors?}}$$

Possible outcomes are all informative. Morgan-only performance would show how far structural information alone can go; improvement from the combined representation would indicate complementary information; no improvement would suggest that the extra fingerprint dimensions do not add sufficiently generalizable signal for this dataset.

6. Expected Next Steps

After the Morgan experiments, the next comparison will use the strongest representation with several model families, potentially including XGBoost, Random Forest, and RBF-SVR, while retaining scaffold-aware evaluation.

A more chemistry-specific graph model such as GINE or a directed message-passing network could also be tested because these architectures explicitly incorporate bond information.

A later extension may use the larger AqSolDB dataset to study how increasing the training-data size affects descriptor-based and learned molecular representations.

Current Takeaway

The central lesson so far is that molecular ML performance depends strongly on both **representation** and **evaluation strategy**. The lowest random split error is not necessarily the most informative number. Scaffold-aware evaluation gives a more difficult but more realistic measure of performance on structurally unfamiliar molecules.

References

[1] John S. Delaney. ESOL: Estimating aqueous solubility directly from molecular structure. *Journal of Chemical Information and Computer Sciences*, 44(3):1000–1005, 2004.

[2] Guy W. Bemis and Mark A. Murcko. The properties of known drugs. 1. molecular frameworks. *Journal of Medicinal Chemistry*, 39(15):2887–2893, 1996.